fibroblast substrate scaffold force microscopy porosity elastic gradient force crosslink polymer traction hydrogel strain viscoelastic microscopy fluorescent hydrogel diffusion strain force modulus network force scaffold force network traction diffusion collagen field viscoelastic substrate porosity modulus stress diffusion adhesion elastic migration gradient elastic diffusion microscopy force polymer rheology porosity strain fibroblast confocal confocal gradient stress fluorescent adhesion fluorescent hydrogel stress density rheology matrix tissue field microscopy modulus crosslink viscoelastic cellular matrix substrate rheology viscoelastic traction microscopy diffusion fibroblast matrix stiffness rheology confocal microscopy hydrogel displacement spheroid microscopy force stress tissue field bioprinting stiffness coating confocal stiffness cellular modulus rheology force polymer field collagen fluorescent scaffold scaffold rheology hydrogel cellular tissue scaffold diffusion displacement collagen strain diffusion displacement viscoelastic stiffness bioprinting network substrate hydrogel adhesion substrate network network granule rheology adhesion bead field granule substrate viscoelastic porosity gradient fibroblast collagen crosslink force confocal diffusion scaffold scaffold scaffold scaffold elastic spheroid scaffold force migration microscopy polymer tissue cellular modulus matrix force elastic granule substrate porosity elastic gradient coating microscopy polymer bioprinting substrate bead stiffness gradient spheroid modulus modulus rheology confocal spheroid spheroid stress hydrogel substrate elastic matrix bead spheroid cellular density coating polymer density gradient substrate porosity coating density stress hydrogel bead density gradient cellular stiffness network porosity porosity crosslink matrix network migration fluorescent scaffold network migration density rheology stiffness coating coating displacement spheroid bead migration stiffness tissue stiffness gradient hydrogel network elastic network spheroid migration matrix polymer spheroid granule spheroid stiffness hydrogel modulus bioprinting migration spheroid adhesion strain matrix hydrogel scaffold confocal scaffold hydrogel cellular cellular collagen coating substrate confocal substrate spheroid stiffness substrate diffusion diffusion collagen coating granule elastic density collagen strain migration polymer coating bead polymer field crosslink fluorescent fibroblast bead porosity viscoelastic collagen force stiffness confocal density viscoelastic crosslink collagen porosity substrate density crosslink coating tissue adhesion granule substrate adhesion substrate spheroid modulus diffusion force fibroblast density density diffusion spheroid elastic diffusion force fluorescent migration displacement traction elastic crosslink tissue diffusion coating microscopy tissue fibroblast crosslink crosslink migration displacement tissue crosslink porosity spheroid crosslink fluorescent density bead diffusion migration tissue collagen viscoelastic modulus scaffold tissue fibroblast microscopy fluorescent strain microscopy polymer stress modulus substrate gradient substrate bead collagen confocal network elastic scaffold rheology cellular network cellular strain crosslink scaffold matrix viscoelastic migration stiffness fibroblast hydrogel gradient coating matrix diffusion confocal tissue coating bioprinting matrix

field crosslink microscopy modulus network elastic hydrogel bead displacement traction adhesion displacement collagen strain bead scaffold substrate porosity

crosslink rheology fibroblast hydrogel displacement force adhesion strain microscopy displacement coating hydrogel bead hydrogel network microscopy bead modulus